



University of the Philippines Cebu

College of Science
Department of Computer Science
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COURSE GUIDE

Course Number	CMSC 178IP
Course Title	Digital Image Processing
Number of Units	3 units (3 hours lecture)
Instructor	Noel Jeffrey Pinton

COURSE OUTLINE

I. Introduction

- a. Overview and Applications
- b. Digital Image Fundamentals
 - i. Image as a $2D$ function $f(x, y)$
 - ii. Pixels, sampling, and quantization
 - iii. Image types and storage formats

II. Image Formation, Representation, and Compression

- a. Image Coding and Compression
 - i. Lossless compression: Huffman, Run-length, LZW
 - ii. Lossy compression: JPEG, JPEG2000, WebP
- b. Imaging Sensors and Systems
 - i. CCD, CMOS, PET, CT
 - ii. Noise sources in image acquisition

III. Image Enhancement

- a. Spatial Domain Methods
 - i. Spatial filtering: smoothing, sharpening
 - ii. Histogram equalization and specification
- b. Frequency Domain Methods
 - i. $2D$ Fourier Transform, frequency representation
 - ii. Low-pass, high-pass, band-pass filtering
- c. Noise Reduction Techniques
 - i. Linear filtering (mean, Gaussian)
 - ii. Nonlinear filtering (median, bilateral)

IV. Image Restoration and Geometric Processing

- a. Image Filtering and Convolution
 - i. Convolution vs. correlation
 - ii. Spatial kernels: blurring, sharpening, edge enhancement
- b. Geometric Transformations
 - i. $2D$ Scaling, rotation, translation, shearing
 - ii. Affine and projective transformations
 - iii. Image registration

V. Feature Extraction and Representation

- a. Edge and Line Detection
 - i. Gradient operators (Sobel, Prewitt, Roberts)
 - ii. Hough transform for lines and circles
- b. Texture and Local Features



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VI. Image Segmentation and Morphology

- a. Segmentation Methods
 - i. Thresholding (global, local, Otsu's method)
 - ii. Region-based segmentation (region growing, watershed)
- b. Morphological Processing
 - i. Binary morphology: erosion, dilation, opening, closing
 - ii. Morphology for grayscale images

VII. Computer Vision and Deep Learning Approaches

- a. Convolutional Neural Networks (CNNs)
 - i. CNN architecture basics: convolution, pooling, fully connected layers
 - ii. Activation functions (ReLU, sigmoid, tanh)
 - iii. Pooling and subsampling
- b. Applications
 - i. Image classification (MNIST, CIFAR-10)
 - ii. Image segmentation
 - iii. Object detection (YOLO, Faster R-CNN)
- c. Generative Models
 - i. Variational Autoencoders (VAE)
 - ii. Generative Adversarial Networks (GAN)

Group Project Guidelines – CMSC 178IP (Digital Image Processing)

1. Group Formation

- Students should organize themselves into **groups of up to 4 members**.
- Smaller groups (2–3) are allowed, but workload expectations remain the same regardless of group size.
- Each group must register their members with the instructor by **Midterms week**.
- Groups are encouraged to form with diverse skills (programming, mathematics, presentation skills, documentation).

2. Project Proposal

- Due by **Midterms**.
- The proposal should include:
 1. Title and objective of the project.
 2. Motivation: why is this problem interesting/important?
 3. Background: relevant references (papers, applications, prior work).
 4. Dataset to be used (must be accessible and ethical to use).
 5. Methodology: which image processing or computer vision techniques will be applied.
 6. Expected outcomes and possible applications.

3. Project Development Process

1. Data Gathering and Preparation

Collect suitable image data (public datasets preferred, or self-collected with consent). Perform preprocessing (resizing, normalization, annotation if needed).



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2. Method Implementation

Implement at least two non-trivial techniques from the course (e.g., morphological processing, edge detection, CNNs). Justify parameter choices (filters, thresholds, architectures).

3. Evaluation and Analysis

Define metrics (PSNR, SSIM, accuracy, F1-score, IoU, etc.). Critically analyze results, including strengths and limitations. Compare against baseline methods if possible.

4. Deliverables

1. **Project Report** with abstract, introduction, related work, methodology, results, discussion, conclusion, and references.
2. **Code Repository** (Python preferred, with README and requirements) that reproduces results.
3. **Oral Presentation** (15–20 minutes during TBD). Each member must speak.

5. Grading Breakdown

- Proposal: 10%
- Implementation & Technical Depth: 30%
- Results & Analysis: 20%
- Report (clarity, structure, rigor): 20%
- Presentation & Delivery: 20%

6. Expectations

- Projects should go beyond toy examples. Examples:
 - Image denoising with classical filters vs. CNN-based denoising.
 - Medical image segmentation using thresholding vs. morphological operations.
 - Object classification using handcrafted features vs. CNN.
- Creativity and originality are highly encouraged.
- Academic integrity must be maintained: no plagiarism, proper citation, and acknowledgment of external code or datasets.